

# README

## Helix OTA — Specification Corpus Index

| Field                 | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Created               | 2026-06-07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Last modified         | 2026-06-07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Status                | active                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Status summary        | <p>Navigable master index of the entire Helix OTA specification corpus rooted at <code>docs/research/main_specs/</code>. Lists every document with a one-line purpose, a relative link, and a status (done / outline / future), grouped by corpus area (00-master foundation, research, additions, 1.0.0-MVP, 1.0.1, and the 1.X future-OS phases). Records how the corpus was produced and what was physically verified.</p> <p>This index is descriptive, not normative: the controlling rules live in <code>00-master/documentation_standards.md</code>. Where a referenced document is itself an outline or future-phase plan, that is reflected in its status column.</p> |
| Status summary detail | <p>Exact HelixConstitution clause numbering (§7.1, §11.4.6, §11.4.29, §11.4.61, §11.4.123) is UNVERIFIED — carried from the corpus convention. There is no standalone 1.0.0-MVP <code>tests/</code> document yet; the four-layer testing strategy currently lives only in the master design (§testing strategy). The six <code>NEW ota-*</code> repositories are specified but not yet created (UNVERIFIED until they exist).</p>                                                                                                                                                                                                                                              |
| Issues                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Issues summary        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Field         | Value                                                                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Index reflects files that physically exist as of Revision 1; planned-but-absent artifacts are marked future.                                                                                                                                                    |
| Fixed         | N/A (initial revision).                                                                                                                                                                                                                                         |
| Fixed summary | Initial corpus index.                                                                                                                                                                                                                                           |
| Continuation  | Add a row when the 1.0.0-MVP tests/ spec is authored; promote the 1.X future-OS outlines to numbered phases; re-run the verification suite (see §8) in CI and link its output; reconcile UNVERIFIED clause numbers against the authoritative HelixConstitution. |

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The Table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED). Every corpus document carries a metadata table followed by a ToC; this index does the same.

## 1. Purpose

This README is the single entry point to the Helix OTA specification corpus. Each row below gives the document's one-line purpose, a relative link from this file, and a status. Use the status column to tell finished specifications apart from outlines and future-phase plans.

Per the anti-bluff convention (HelixConstitution §7.1 / §11.4.6, UNVERIFIED), nothing in this index is fabricated: every linked file physically exists in the corpus, every status reflects the linked document's own metadata, and any claim not confirmed from a real source is tagged UNVERIFIED. The full rule set lives in [00-master/documentation\\_standards.md](#) §8.

## 2. How this corpus was produced

The corpus was produced in three stages:

1. **Multi-agent research** — parallel research waves surveyed the OTA landscape and produced the twelve stack notes in research/stacks/ plus the landscape report, then reconciled the two operator input drafts in additions/ into research/additions\_synthesis.md.
2. **Adversarial review** — every contradiction (between the drafts, against the locked decisions, and against the HelixConstitution) was surfaced and resolved, and the evidence-backed choices were captured as the five ADRs in research/adr/. The master design and the threat model consolidate the surviving decisions, with the requirements traceability matrix acting as the completeness check that no operator requirement was dropped.
3. **Validation** — the executable/parseable 1.0.0-MVP artifacts (OpenAPI, SQL migrations, Kubernetes manifests, docker-compose) were run through real validators on the build host, and the export pipeline was exercised to render the corpus. The captured commands and outputs are recorded in 1.0.0-mvp/VALIDATION\_EVIDENCE.md and 00-master/export\_pipeline.md. See §8.

## 3. Status legend

| Status  | Meaning                                                                                 |
|---------|-----------------------------------------------------------------------------------------|
| done    | Authored to full corpus depth; for executable artifacts, physically validated (see §8). |
| outline | Authored as a structured plan/outline; depth follows a completed phase.                 |
| future  | Forward-looking research outline for a not-yet-scheduled phase.                         |

## 4. 00-master — foundation

Foundation documents that anchor the whole corpus. Directory: 00-master/.

| Document                           | Purpose                                                                                                                                                                                                                   | Status |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <u>Master design specification</u> | Canonical master design: vision, locked decisions, architecture, mandated stack, MVP scope, security/trust model, data model, rollout engine, telemetry, submodule map, phased layout, testing strategy, execution model. | done   |
| <u>Glossary</u>                    | Canonical definitions of domain terms used across                                                                                                                                                                         | done   |

| Document                                | Purpose                                                                                                                                                                   | Status |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <u>Threat model</u>                     | the corpus, each with where it appears.<br>STRIDE-based threat model for the 1.0.0-MVP: twelve OTA threats with attack, impact, MVP mitigation, residual risk.            | done   |
| <u>Submodule reuse map</u>              | Catalogue-first reuse/extend/new map binding each component to verified submodules; specifies the six NEW ota-* modules.                                                  | done   |
| <u>Documentation standards</u>          | Normative rules: mandatory metadata table, ToC, file naming, export targets, diagram conventions, cross-references, anti-bluff/UNVERIFIED, canonical submodule catalogue. | done   |
| <u>Requirements traceability matrix</u> | Every operator requirement mapped to phase × component/spec × status (specified / researched / pending).                                                                  | done   |
| <u>Export pipeline</u>                  | The real, tested Markdown→HTML/DOCX(/PDF) + Mermaid→SVG/PNG pipeline (scripts/export_docs.sh) with captured live-run output.                                              | done   |
| <u>Diagram renders (_exports/)</u>      | Generated artifacts: master-design diagram-01 in .mmd/.svg/.png, plus rendered .html and .docx of the master design.                                                      | done   |
| ADR index                               | Architecture decisions ADR-0001..0005 live in <u>research/adr/</u> and are indexed in the master design §ADRs and in §5 below.                                            | done   |

## 5. research — landscape, stack notes, ADRs, synthesis

Directory: [research/](#).

| Document                             | Purpose                                                                                                                                         | Status |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <a href="#">OTA landscape report</a> | Landscape survey and engine-selection synthesis across the researched OTA stacks.                                                               | done   |
| <a href="#">Additions synthesis</a>  | Reconciles the two operator drafts: extracts reusable elements, maps each to a destination, records every contradiction with a resolution rule. | done   |

Stack research notes — directory [research/stacks/](#) (12 notes):

| Note                                          | Purpose                                                          | Status |
|-----------------------------------------------|------------------------------------------------------------------|--------|
| <a href="#">Android AVB / rollback</a>        | Android Verified Boot, dm-verity, boot_control, rollback.        | done   |
| <a href="#">Android update_engine API</a>     | update_engine API surface and OTA package structure.             | done   |
| <a href="#">Android 15 Virtual A/B</a>        | Android 15 Virtual A/B with compression — device-side core.      | done   |
| <a href="#">AOSP update_engine</a>            | AOSP update_engine / Android OTA mechanism.                      | done   |
| <a href="#">Commercial / OSS fleet survey</a> | balena, Toradex Torizon, Memfault, Foundries.io fleet platforms. | done   |
| <a href="#">Eclipse hawkBit</a>               | hawkBit rollout management server.                               | done   |
| <a href="#">Mender</a>                        | Mender update client/server.                                     | done   |
| <a href="#">OSTree</a>                        | OSTree / libostree / rpm-ostree atomic file-based updates.       | done   |
| <a href="#">RAUC</a>                          | RAUC fail-safe A/B bundle updater.                               | done   |
| <a href="#">SWUpdate</a>                      | SWUpdate embedded Linux .swu updater.                            | done   |
| <a href="#">TUF + go-tuf/v2</a>               |                                                                  | done   |

| Note          | Purpose                                                                                | Status |
|---------------|----------------------------------------------------------------------------------------|--------|
| <u>Uptane</u> | TUF supply-chain integrity with go-tuf/v2.<br>Uptane automotive-grade trust framework. | done   |

Architecture Decision Records — directory [research/adr/](#) (5 ADRs, status Proposed):

| ADR                                | Purpose                                                | Status          |
|------------------------------------|--------------------------------------------------------|-----------------|
| <u>ADR-0001 Wrapped engine</u>     | hawkBit vs Mender vs AOSP-native-only.                 | done (Proposed) |
| <u>ADR-0002 Supply-chain trust</u> | Plain signing vs TUF vs Uptane, and MVP timing.        | done (Proposed) |
| <u>ADR-0003 Server topology</u>    | Modular monolith vs microservices, with scale trigger. | done (Proposed) |
| <u>ADR-0004 Transport</u>          | HTTP/3 (QUIC) + Brotli with HTTP/2 fallback.           | done (Proposed) |
| <u>ADR-0005 Delta updates</u>      | AOSP block diffs vs custom, and phase placement.       | done (Proposed) |

## 6. additions — operator inputs

Raw operator-supplied input drafts, preserved as the source for the synthesis (§5). Directory: [additions/](#).

| Document                          | Purpose                                                  | Status                |
|-----------------------------------|----------------------------------------------------------|-----------------------|
| <u>Initial research (draft 1)</u> | First operator draft of the Helix OTA plan and research. | done (operator input) |
| <u>Initial research (draft 2)</u> | Second operator draft (parallel-wave research framing).  | done (operator input) |

## 7. Phase specifications

### 7.1 1.0.0-mvp

The first shippable phase: native Android 15 A/B on device, a Go + Gin control plane, a React dashboard, and a PostgreSQL + MinIO/S3 artifact store. Directory: [1.0.0-mvp/](#).

API — [1.0.0-mvp/api/](#):

| Document                                | Purpose                                                                   | Status           |
|-----------------------------------------|---------------------------------------------------------------------------|------------------|
| <a href="#">Endpoint specification</a>  | The /api/v1 REST endpoint specification.                                  | done             |
| <a href="#">OpenAPI 3.1 description</a> | Machine-readable OpenAPI 3.1 (12 paths, 24 schemas); lint-validated (§8). | done (validated) |

Database — [1.0.0-mvp/database/](#):

| Document                             | Purpose                                                                 | Status           |
|--------------------------------------|-------------------------------------------------------------------------|------------------|
| <a href="#">Schema</a>               | The 1.0.0-MVP PostgreSQL schema.                                        | done             |
| <a href="#">Migrations (up/down)</a> | 001_initial_schema up + down SQL; applied against a live Postgres (§8). | done (validated) |

Security — [1.0.0-mvp/security/](#):

| Document                                   | Purpose                                                   | Status |
|--------------------------------------------|-----------------------------------------------------------|--------|
| <a href="#">Key management</a>             | MVP signing-key lifecycle and custody.                    | done   |
| <a href="#">Signing &amp; verification</a> | Artifact signing and verification flow.                   | done   |
| <a href="#">Transport security</a>         | TLS/transport security for the control plane and clients. | done   |

Server — [1.0.0-mvp/server/](#):

| Document                             | Purpose                                 | Status |
|--------------------------------------|-----------------------------------------|--------|
| <a href="#">Architecture</a>         | Modular-monolith server architecture.   | done   |
| <a href="#">Artifact validation</a>  | Artifact upload validation pipeline.    | done   |
| <a href="#">Telemetry processing</a> | Telemetry processing and device health. | done   |

Android client — [1.0.0-mvp/client\\_android/](#):

| Document                                  | Purpose                            | Status |
|-------------------------------------------|------------------------------------|--------|
| <a href="#">Integration guide</a>         | Android agent integration guide.   | done   |
| <a href="#">Build integration</a>         | Android agent build integration.   | done   |
| <a href="#">update_engine integration</a> | Apply path via AOSP update_engine. | done   |

| Document             | Purpose                                                                                                         | Status                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <u>Code snippets</u> | Reference Kotlin/AOSP build snippets (downloader, verifier, applier, poll worker, telemetry, Android.bp, etc.). | done (snippets; compile pending toolchain) |

Deployment — 1.0.0-mvp/deployment/:

| Document                    | Purpose                                                                                       | Status           |
|-----------------------------|-----------------------------------------------------------------------------------------------|------------------|
| <u>Overview</u>             | Deployment overview for the MVP.                                                              | done             |
| <u>MinIO setup</u>          | MinIO/S3 artifact-store setup.                                                                | done             |
| <u>docker-compose</u>       | Single-host MVP compose stack.                                                                | done             |
| <u>Kubernetes manifests</u> | namespace, ota-server deployment + service, postgres statefulset; kubeconform-validated (§8). | done (validated) |

Cross-cutting:

| Document                   | Purpose                                                                                                                        | Status |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|
| <u>VALIDATION EVIDENCE</u> | Physical, reproducible validation of the MVP artifacts with real tools — commands + captured output.                           | done   |
| Tests                      | Four-layer testing strategy is currently only in the <u>master design</u> ; no standalone tests/ spec exists yet (UNVERIFIED). | future |

## 7.2 1.0.1-staged-rollout

| Document                | Purpose                                                                                              | Status  |
|-------------------------|------------------------------------------------------------------------------------------------------|---------|
| <u>Phase 1.0.1 plan</u> | First post-MVP phase: staged rollout, health-gated halt, device-side trust (TUF), end-user rollback. | outline |

## 7.3 1.X future-OS phases

Each extends Helix OTA to a new OS family via the OS-adapter seam; the control plane is reused unchanged and only a device adapter is added.

| Document            | Purpose                                                                    | Status |
|---------------------|----------------------------------------------------------------------------|--------|
| <u>1.X Linux</u>    | Linux support (RAUC / OSTree-bootc / SWUpdate / package-manager backends). | future |
| <u>1.X Windows</u>  | Windows 10/11, IoT/LTSC, Server support.                                   | future |
| <u>1.X Other OS</u> | macOS, *BSD, RTOS, and other upstream OSes.                                | future |

## 8. Verification evidence

The following were physically executed; do not treat as claims. Full commands and captured output are in 1.0.0-mvp/VALIDATION\_EVIDENCE.md and 00-master/export\_pipeline.md.

- **Live PostgreSQL migration apply** — both `001_initial_schema` up + down migrations applied against a live server with `psql -v ON_ERROR_STOP=1` (verified).
- **OpenAPI lint** — `npx @redocly/cli lint api/openapi.yaml → valid` (1 cosmetic info-license warning); 12 paths, 24 component schemas (verified).
- **kubeconform** — `kubeconform -summary -strict deployment/ kubernetes/*.yaml` over the four manifests (verified).
- **Export render** — `scripts/export_docs.sh` rendered Markdown→HTML/DOCX and Mermaid→SVG/PNG (pandoc + mmdc present); PDF requires a LaTeX engine that was absent on the build host, so PDF is deferred to a CI container (UNVERIFIED) rather than faked.

Validations requiring tools absent on the build host (docker, yq, LaTeX, drawio, plantuml, dot) are deferred to a CI container, not faked (UNVERIFIED until that job runs).

## 9. Submodule reference (catalogue + the six NEW ota-\* modules)

This index references only real catalogue submodules and the six NEW ota-\* modules. The canonical catalogue names are fixed in 00-master/documentation\_standards.md §9; the bindings (reuse / extend / new) are in 00-master/submodule\_reuse\_map.md.

The six NEW modules to be created (specified, not yet created — UNVERIFIED until they exist):

| <b>NEW module</b>        | <b>Purpose</b>                                                                     |
|--------------------------|------------------------------------------------------------------------------------|
| ota-protocol             | Shared wire types, manifest schema, status/event enums (Go + KMP); pure contracts. |
| ota-artifact-validator   | Structure / hash / signature / metadata validation pipeline; no transport.         |
| ota-rollout-engine       | Staged-rollout + halt/advance logic, OS-agnostic; no HTTP.                         |
| ota-update-engine-bridge | Android-only wrapper over AOSP update_engine / boot_control.                       |
| ota-android-agent        | KMP device agent: poll, download, verify, apply, report.                           |
| ota-telemetry-schema     | Telemetry event / metric schema + codecs; shared server + agent.                   |